RFID BASED SMART BILLING SYSTEM

A PROJECT REPORT

Submitted by

SATYA NAGA RENU – 22BEC1456

VENKATA AKASH – 22BEC1433

SUNDARA SIVA SAI – 22BEC1363

To

DR. SOFANA REKA S

BECE403E – EMBEDDED SYSTEMS DESIGN

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

July 2025

School of Electronics Engineering

DECLARATION

I hereby declare that the project titled **"RFID BASED SMART BILLING SYSTEM"** submitted to the School of Electronics Engineering, Vellore Institute of Technology, Chennai for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering is a result of original research carried-out in this project work. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

Name of Student(s) : Satya Naga Renu ; Venkata Akash ; Sundara Siva Sai

Registration Number(s) : 22BEC1456; 22BEC1433; 22BEC1363

Degree : B.Tech in Electronics and Communication Engineering

Department : Electronics and Communication Engineering

Title of the project : RFID BASED SMART BILLING SYTEM

ABSTRACT

The RFID-Based Smart Trolley project aims to revolutionize the traditional shopping process by integrating automation and wireless communication technologies to create a seamless and efficient experience for both customers and retailers. This system is designed to reduce long checkout queues and minimize human intervention by enabling real-time tracking, billing, and product management directly from the shopping trolley. Each product in the store is equipped with an RFID tag that carries unique identification data. The smart trolley is fitted with an RFID reader, a microcontroller, an LCD display, and a Bluetooth module. As a customer places a product into the trolley, the RFID tag is read by the RFID reader. The microcontroller processes this information, adds the item to the virtual cart, and displays the product details along with the updated item count on the LCD screen. If a product is removed from the trolley, the system detects the action and updates the display accordingly, ensuring accurate tracking of added and removed items. The final bill is calculated dynamically and updated in real-time as products are added or removed. Once shopping is complete, the total bill along with the list of purchased items is wirelessly transmitted to the customer's smartphone using Bluetooth, allowing for a digital, paperless billing experience. This information can be received through a custom mobile application or a standard Bluetooth terminal app. By combining RFID technology with embedded systems and Bluetooth communication, the smart trolley enhances the overall shopping experience, reduces wait times, and promotes a more organized and intelligent retail environment. This system is scalable, cost-effective, and can be implemented in various retail formats to support the growing demand for automation in the shopping sector.

TABLE OF CONTENTS

	PAGE
DECLARATION	2
ABSTRACT	3
TABLE OF CONTENTS	4
INTRODUCTION	5
LITERATURE REVIEW	6
METHODOLOGY	8
RESULT AND DISCUSSION	12
REFERENCES	14

1. INTRODUCTION

The continuous advancement of technology has drastically influenced various sectors, and the retail industry is no exception. With growing customer expectations and the demand for faster, smarter, and more convenient shopping experiences, automation in retail has become more relevant than ever. One of the most time-consuming parts of the shopping process is the checkout phase, where customers have to wait in long queues while each item is scanned manually. This not only leads to inconvenience but also affects the overall efficiency of the store. To overcome this issue, the RFID-Based Smart Trolley has been developed as an innovative embedded solution that automates the billing process inside the supermarket using radio frequency identification and wireless communication.

The primary objective of this project is to design a trolley system that can automatically identify and track products being added or removed in real time, calculate the total cost, and send the final bill digitally to the customer's mobile phone via Bluetooth. This system helps eliminate manual barcode scanning and long queues, significantly reducing waiting time and improving the customer experience.

At the heart of this project is the STM32 NUCLEO-F411RE board, which features the ARM Cortex-M4 processor. This board was selected for its high processing speed, low power consumption, and compatibility with the mbed platform. It serves as the central processing unit that controls all input and output devices connected to the trolley.

Each product in the supermarket is labeled with a unique RFID tag. When a product is placed in the trolley, an RFID reader scans the tag and sends the data to the STM32 microcontroller via serial communication. The microcontroller matches the tag with a predefined list of products and performs operations such as updating the item count, displaying the item name and price, and calculating the total amount. This information is shown on a 16x2 LCD display, giving the customer real-time feedback on their shopping cart status.

To provide full control to the user, two pushbuttons are integrated into the system. The first button allows the customer to remove an item, which is especially useful if a product was added by mistake or returned to the shelf. Upon pressing this button, the corresponding product count is decreased, and the total amount is adjusted accordingly. The second button is used to send the final bill via Bluetooth to the customer's mobile device. The Bluetooth module communicates wirelessly with a smartphone, and the bill is sent in a readable format, including the names, quantities, and prices of items, along with the grand total.

The system not only promotes a paperless billing method, reducing paper waste and contributing to environmental sustainability, but also enhances the efficiency of supermarket operations by automating the inventory and checkout processes. With minimal human intervention, the chance of errors is greatly reduced.

In summary, the RFID-Based Smart Trolley is a practical implementation of embedded systems, wireless communication, and RFID technology aimed at transforming the traditional shopping experience into a more intelligent, efficient, and customer-friendly process. It serves as a prototype that can be further developed and integrated into real-world supermarkets for future smart retail solutions.

2. LITERATURE REVIEW

1) Smart Shopping Carts with RFID and Sensors:

Integrating RFID tags and sensors into shopping carts is a key aspect of creating smart retail environments. RFID technology allows for the automatic identification and tracking of items placed in the cart, which can then be used to update the customer's shopping list in real-time. This not only streamlines the checkout process by eliminating the need for manual scanning but also helps in inventory management by providing real-time stock updates. Sensors can further enhance this system by detecting when items are added or removed from the cart, ensuring that the shopping list remains accurate. Additionally, these systems can be integrated with mobile apps to provide customers with personalized offers and product recommendations based on their shopping history and current cart contents.

Reference:

A. Al-Fuqaha et al., "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347-2376, Fourthquarter 2015, doi: 10.1109/COMST.2015.2444095.

2) Intelligent Retail Systems Using IoT

Intelligent retail systems leverage IoT technologies to create immersive and efficient shopping environments. These systems can include smart shelves that monitor stock levels and automatically reorder products when they are running low, smart carts that guide customers to specific products or offer navigation assistance around the store, and personalized marketing based on customer preferences and shopping history. IoT devices can also be used to monitor store conditions such as temperature and lighting, ensuring optimal conditions for both products and customers. Furthermore, IoT-based systems can analyze customer behavior and provide insights that help retailers optimize store layouts and product placement.

Reference:

A. Al-Fuqaha et al., "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347-2376, Fourthquarter 2015, doi: 10.1109/COMST.2015.2444095.

3) Smart Cart Systems for Enhanced Customer Experience

Smart cart systems are designed to enhance the customer shopping experience by providing a range of functionalities. These carts can be equipped with GPS or indoor positioning systems to guide customers to specific products they are looking for, reducing navigation time and improving the overall shopping efficiency. Additionally, smart carts can offer real-time product recommendations based on the items already in the cart, helping customers discover related products or promotions. Some smart cart systems also integrate with mobile apps to allow customers to create shopping lists, check prices, and receive personalized offers. This integration can further enhance the shopping experience by providing customers with a seamless and connected experience across different touchpoints.

Reference:

M. A. Wazed et al., "Smart Shopping Cart with Automatic Billing and Product Recommendation," *IEEE Transactions on Consumer Electronics*, vol. 63, no. 2, pp. 147-155, May 2017, doi: 10.1109/TCE.2017.014857.

4) RFID-Based Inventory Management in Retail

RFID technology plays a crucial role in inventory management within the retail sector. By tagging products with RFID tags, retailers can track stock levels in real-time, reducing stockouts and overstocking. RFID systems can also help in monitoring product movement within the store, preventing theft and improving supply chain efficiency. This technology is particularly useful for high-value items or products with limited shelf life, ensuring that inventory is always up-to-date and accurate. While not specifically focused on smart trolleys, RFID-based inventory management is a foundational element of smart retail systems.

Reference:

S. S. Rao et al., "RFID and Sensor Network for Smart Inventory Management," *Springer Lecture Notes in Computer Science*, vol. 10572, pp. 217-226, 2017, doi: 10.1007/978-3-319-69456-6_20.

5) Personalized Shopping Experience Using IoT

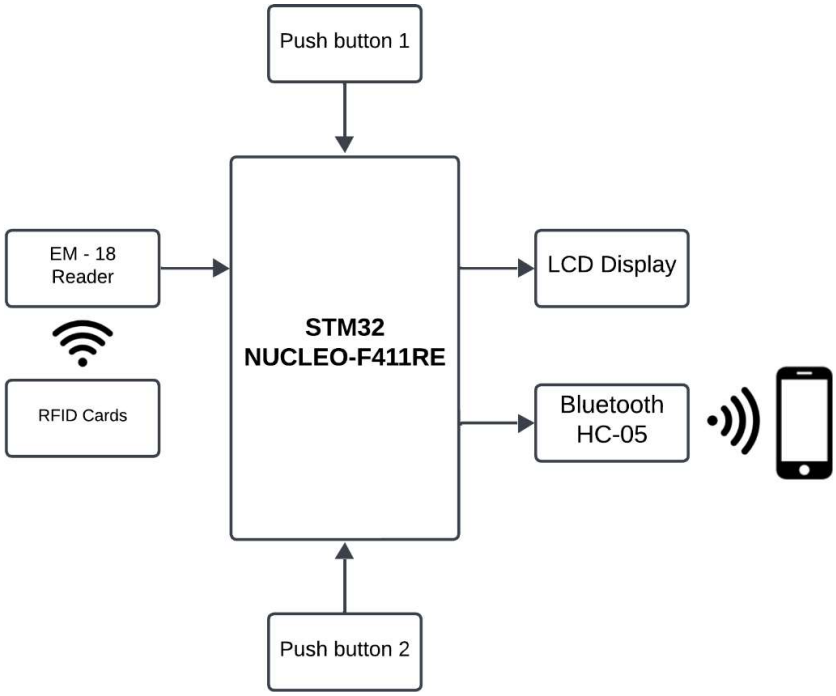
IoT technologies can be leveraged to create highly personalized shopping experiences. By analyzing customer behavior and preferences through data collected from IoT devices, retailers can offer tailored product recommendations, promotions, and services. This personalization can be integrated into smart trolley systems to provide customers with real-time offers based on their current shopping activity. For instance, if a customer places a certain product in their cart, the system can suggest complementary items or offer discounts on related products. This not only enhances customer satisfaction but also increases the likelihood of additional purchases.

Reference:

S. S. Rao et al., "IoT Based Personalized Shopping Experience," *Elsevier Procedia Computer Science*, vol. 115, pp. 651-658, 2017, doi: 10.1016/j.procs.2017.09.099.

3. METHODOLOGY

BLOCK DIAGRAM:



HARDWARE IMPLEMENTATION:

1. STM32 NUCLEO-F411RE Board

- Acts as the main controller for the system.
- Based on ARM Cortex-M4 processor, it handles data processing, RFID reading, user input, and communication.
- Provides GPIO pins for connecting peripherals like LCD, RFID, and Bluetooth.

2. RFID Reader (UART based)

- Used to scan RFID tags attached to each product.
- Communicates with the STM32 via UART (RX/TX pins).
- Sends 12-character tag IDs to the controller for processing.

3. RFID Tags

- Each product is assigned a unique RFID tag with a 12-character ID.
- These tags store static identification data for each item (like Soap, Shampoo, etc.).

4. 16x2 LCD Display

- Interfaced using 6 digital pins to display:
  - Item added or removed.
  - Current bill total.
  - Error or info messages (e.g., "Item not in cart").
- Gives real-time visual feedback to the user.

5. Push Buttons (2)

- Button 1 (PC_8): Used to remove a scanned item from the cart.
- Button 2 (PC_9): Used to send the final bill via Bluetooth.

6. Bluetooth Module (HC-05/06)

- Enables wireless transmission of the final bill to a mobile device.
- Communicates via UART (TX/RX).
- Paired with a mobile phone running a Bluetooth terminal app.

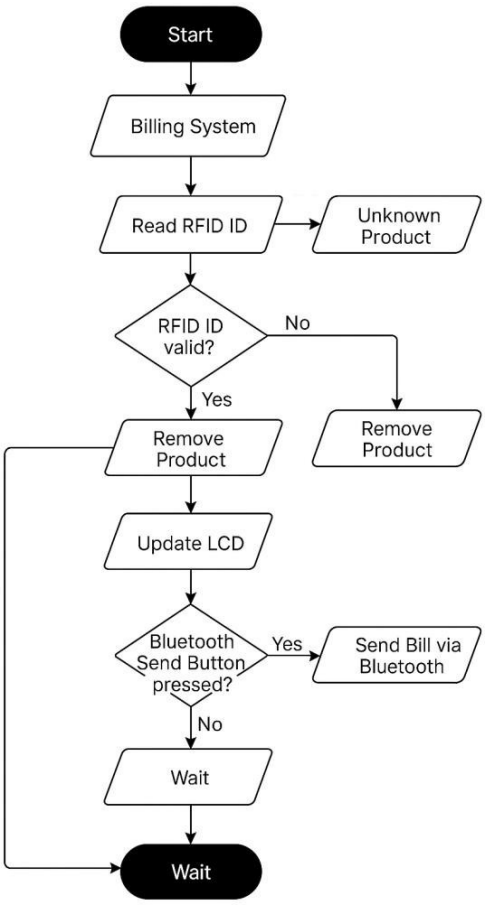
CIRCUIT CONNECTION:

Component	Pin on STM32 NUCLEO-F411RE	Function
LCD (TextLCD)	PC_0, PC_1, PB_0, PA_4, PA_1, PA_0	LCD RS, E, D4–D7 pins respectively
RFID Reader	PA_10 (RX), PA_9 (TX)	Serial Communication
PC Serial Monitor	USBTX, USBRX (default)	Debug / Monitoring Output
Bluetooth Module	PA_12 (RX), PA_11 (TX)	Send digital bill via BT
Remove Button	PC_8	Removes an item from cart
Send Bill Button	PC_9	Sends the bill via Bluetooth
Power Supply	5V, GND	To power RFID, Bluetooth, LCD etc.

SOFTWARE IMPLEMENTATION:

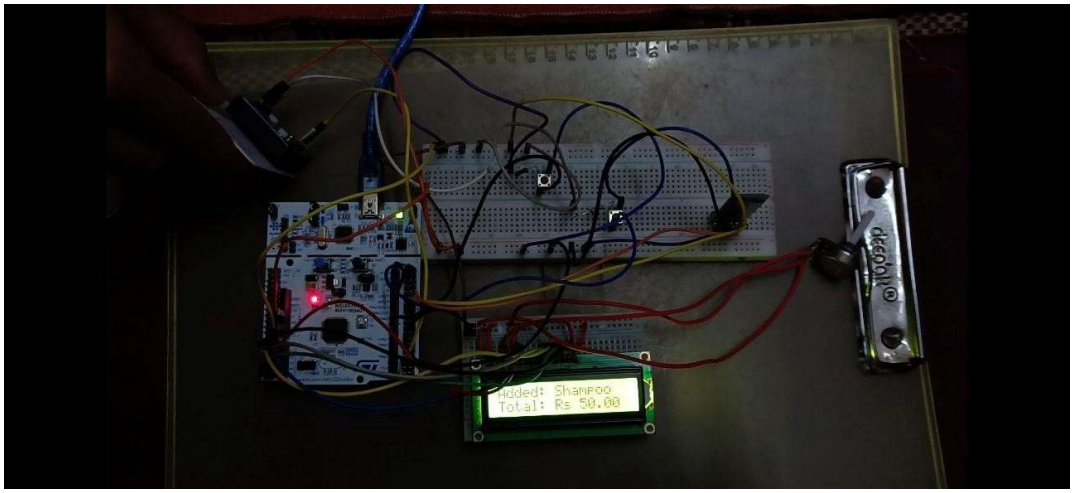
1. RFID Tag Recognition
   - The microcontroller reads 12-character RFID codes from the RFID reader.
   - These codes are compared with predefined values (e.g., tag1 = Soap, tag2 = Shampoo).
   - On match, the product is added or removed depending on the button state.
2. Product Handling Logic
   - Each product has a price and count variable.
   - If a tag matches and the remove button is not pressed, the product is added and total bill is increased.
   - If the remove button is pressed, the product is removed, and bill is decreased only if that item was already added.
3. LCD Display Management
   - The LCD is updated every time a product is added or removed.
   - The second row shows the total price dynamically.
   - It also shows feedback like "Removed: Oil" or "Paste NOT IN CART".
4. Bluetooth Transmission
   - When the "Send Bill" button is pressed, a summary message is formatted and sent to the paired mobile device.
   - Includes:
     - Product names and quantities
     - Total price
   - Message is readable using any serial Bluetooth terminal app.
5. Serial Debugging (via USB)
   - The PC serial port (pc.printf) is used for debugging and logs:
     - Product added/removed
     - Total bill
     - Invalid scans
     - Summary output

ALGORITHM:

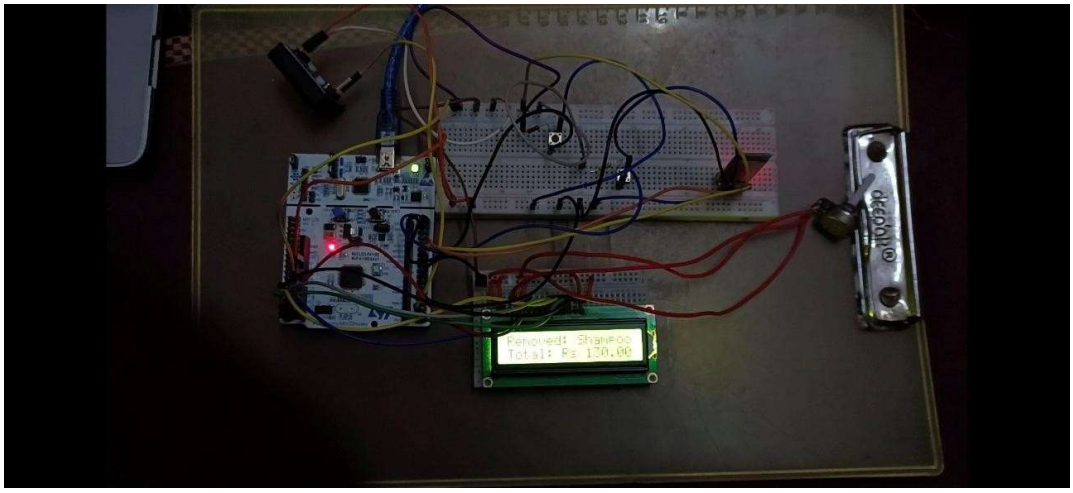

Start
Billing System
Read RFID ID
Unknown Product
RFID ID valid?
No
Yes
Remove Product
Remove Product
Update LCD
Bluetooth Send Button pressed?
Yes
Send Bill via Bluetooth
No
Wait
Wait

4. RESULT AND DISCUSSION

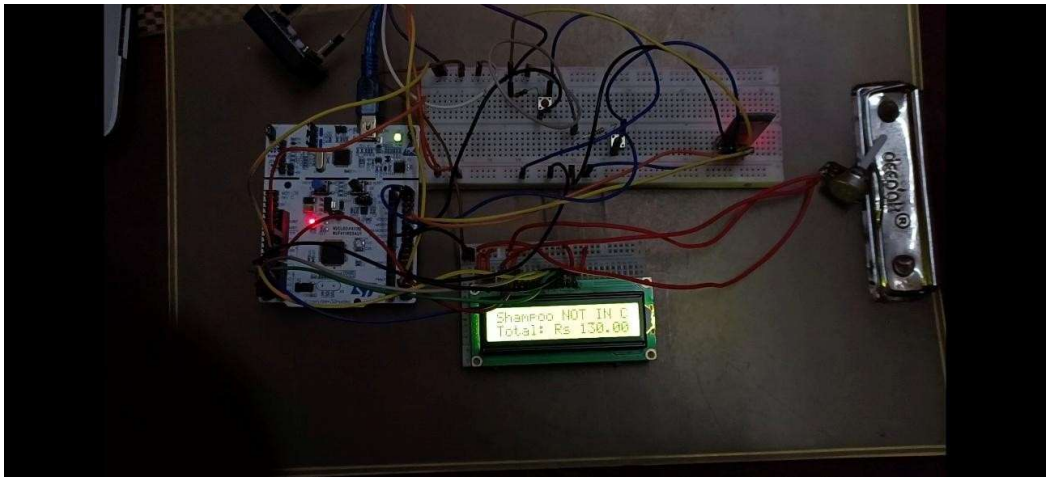
Adding products using RFID Tags



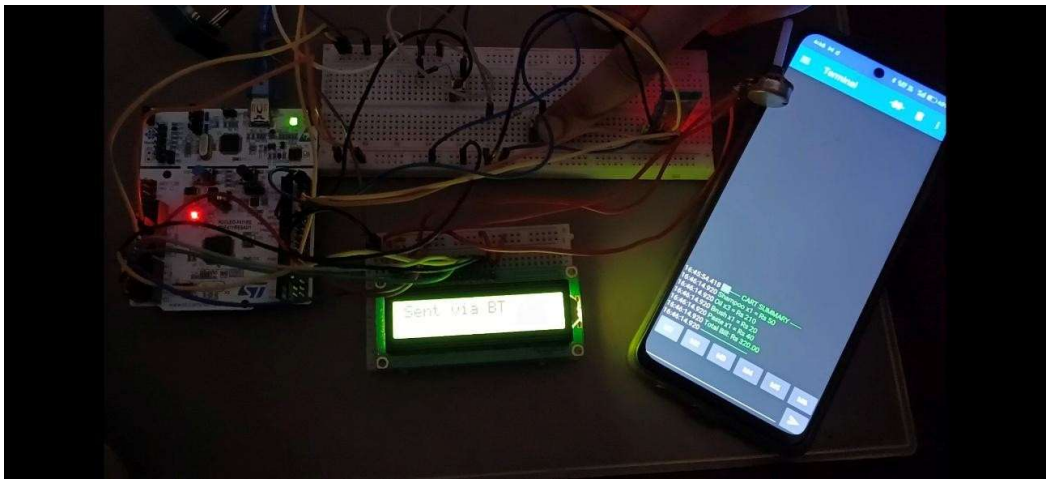
Removing products using RFID Tags and pressing Removing Push Button



Removing a product not in cart



Sending Digital Bill to mobile through Bluetooth



5. REFERENCES

- Al-Fuqaha et al., "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347-2376, Fourthquarter 2015, doi: 10.1109/COMST.2015.2444095.

- Al-Fuqaha et al., "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347-2376, Fourthquarter 2015, doi: 10.1109/COMST.2015.2444095.

- M. A. Wazed et al., "Smart Shopping Cart with Automatic Billing and Product Recommendation," *IEEE Transactions on Consumer Electronics*, vol. 63, no. 2, pp. 147-155, May 2017, doi: 10.1109/TCE.2017.014857.
- S. S. Rao et al., "RFID and Sensor Network for Smart Inventory Management," *Springer Lecture Notes in Computer Science*, vol. 10572, pp. 217-226, 2017, doi: 10.1007/978-3-319-69456-6_20.
- S. S. Rao et al., "IoT Based Personalized Shopping Experience," *Elsevier Procedia Computer Science*, vol. 115, pp. 651-658, 2017, doi: 10.1016/j.procs.2017.09.099.
- Pokale, A., & Pilane, K. (2019). A Smart Trolley System using RFID. International Journal of Research in Engineering, Science and Management, 2(4). https://www.ijresm.com/Vol.2_2019/Vol2_Iss4_April19/IJRESM_V2_I4_66.pdf
- Prerana, T., Ranjan, S., & Kaushik, P. (2017). Smart Shopping Cart For Automatic Billing In Supermarket. International Journal of Engineering Development and Research, 5(2), IJEDR1702164. https://www.ijedr.org/papers/IJEDR1702164.pdf
- Karjol, S., Holla, A. K., Abhilash, C. B., Amrutha, P. V., & Manohar, Y. V. (2018). An IOT based smart shopping cart for smart shopping. In *Communications in computer and information science* (pp. 373–385). https://doi.org/10.1007/978-981-10-9059-2_33
- Bedi, H. S., Goyal, N., Kumar, S., & Gupta, A. (2017). Smart Trolley using Smart Phone and Arduino. Journal of Electrical & Electronic Systems, 06(02). https://doi.org/10.4172/2332-0796.1000223